eedi-mining-misconceptions-in-mathematics

Private 11th (Public 9th) Place Solution Summary

Rank	11
Team	H&Y&L
Author	huangxiaojian
Topic ID	551424
Version	Original

Source: https://www.kaggle.com/competitions/eedi-mining-misconceptions-in-mathematics/discussion/551424
Retrieved with Kaggle CLI on 2026-10-21.

Firstly, thanks to Kaggle organizers and my team mates.
I will briefly share our solution.

Overview

Our pipeline mainly follows flagembedding https://github.com/FlagOpen/FlagEmbedding. It can be divided into 2 stage: retrieval and reranker.

@sayoulala : https://www.kaggle.com/competitions/eedi-mining-misconceptions-in-mathematics/discussion/543519 Thanks for sharing, that's where a lot of our design of pipeline comes from.

Retrieval

In addition to training data, we also use external datas from the disscussion https://www.kaggle.com/competitions/eedi-mining-misconceptions-in-mathematics/discussion/533764 and https://www.kaggle.com/competitions/eedi-mining-misconceptions-in-mathematics/discussion/541222, thanks for sharing. For convenience, we refer to them as ext1 and ext2.

We divide the training process into 3 stage. First, we train the model using ext1, then we train the model using ext2, finally we train using training data.

We observe that, maybe the ext1 has noise, so when we train the model using ext1+training data, the performance is worse than 2-stage training. And for ext2, because it lacks the construct name and subject name, we continue using the multi-stage training strategy.

Reranker

We obtain three trained retrieval models from each of the above three stages. And then We use these three models to create hard negative data for ext1, ext2, and training data, respectively. Then we also divide the training process of reranker into 3 stages, in each stage we use corresponding hn data to train.

Some Details

When we train the 3rd stage of retrieval, we use negative cross device to enlarge the number of negatives for each positive. When we use it, public score 0.567->0.574.

We use Qwen2.5-32B for retrieval and reranker. Because of the limit, we only rerank top 18 in inference.

Model	Public LB
retrieval	0.582
retrieval+reranker	0.623

Code

The main training code is provided in https://github.com/MilchstraB/Eedi

Supplemental Q&A

> **no fit just luck · 2024-12-13T07:30:51.807000**
>
> It's good to see a simple but effective method. Especially, by using two open source datasets can achieve such a nice LB and PB. May I ask how do you use MalAlgoQA, since there is no misconception in it. It can be observed that synthetic data is import for the competition, especially the extra data should cover all misconceptions. However, it is hard to find this point in your sharing method. May you sharing more about the synthetic data you used, since it may be the main point for get such a great PB.

> **huangxiaojian · 2024-12-13T07:41:19.657000**
>
> For MalAlgoQA, I just treat the rationale of incorrect answer as misconception. In my experiments, the external datas can bring about 0.02~0.03 score improvement in retrieval stage. I think the data of high quality is most important, because the main gain comes from parameter adjustments when training on the training set, like number of negatives or batch size.

> **no fit just luck · 2024-12-13T08:19:35.717000**
>
> Great job with such two dataset can get retrieval model for LB 0.582. We also tried using the first dataset you mentioned, but it does not work. And maybe MalAlgoQA is the salient reason, although it guide model on predicting different misconceptions, which usually is not useful for improving model performance.

> **huangxiaojian · 2024-12-13T08:33:51.580000**
>
> We first using ext1 to train retrieval. The model we used for training is Qwen-2.5-7B-math. Then we using the trained model to relabel ext1, which caculates the similarity of ext1 and misconceptions the competition offered.

> **no fit just luck · 2024-12-13T08:57:49.097000**
>
> May I ask more clearly. You mean you used ext1 dataset trained Qwen-2.5-7B-math, and then used the model relabel ext1, in order to correct the original misconception in ext1? It's a method we have not considered before, learn a lot from it.

> **huangxiaojian · 2024-12-13T09:01:08.677000**
>
> Yes, we follow the 3-stage training strategy to train Qwen-2.5-7B-math and then relabel ext1.